

Gas Discharge Arc Lamp Manufacturers — 528 nm SPR / PGM-REE Processing

Specification: Dielectric breakdown gas discharge apparatus — gaseous medium in dielectric housing sustaining electrical arc discharge — primary emission 528 nm (green) for Au/Ag SPR excitation in comminuted ore processing — continuous or pulsed UV-Vis-NIR output across PGM and REE absorption bands.

Technology Overview

Technology	Gas Medium	Housing	528 nm?	UV	NIR
Xe Short Arc	Xenon	Quartz (SiO ₂)	Continuum ✓	< 200 nm	2500 nm
Hg-Xe Arc	Hg + Xe	Quartz	+ 546nm line	185 nm+	2500 nm
Kr Arc	Krypton	Quartz	Continuum ✓	< 200 nm	1000 nm
LDLS (laser Xe)	Xenon	Quartz dielectric	Continuum ✓	170 nm	2100 nm
CERMAX	Xenon	Al ₂ O ₃ ceramic	Continuum ✓	< 250 nm	1000 nm
DBD excimer	Xe/Kr/Hg	Dielectric glass	Narrow UV only	VUV	limited
Microwave plasma	Sulfur vapor	Quartz	~520 nm ★	visible	limited

★★★★★ BEST MATCH — Energetiq Technology (Hamamatsu group)

Product: LDLS EQ-99XFC / EQ-77

Laser-Driven Light Source: a CW laser sustains a Xe plasma inside a sealed fused-quartz dielectric bulb — electrodeless, no electrode erosion. Delivers 170–2100 nm continuous output covering every PGM and REE absorption band. ~10× higher radiance than conventional Xe arcs. Fiber-coupled output direct to reaction vessel.

Website: <https://www.energetiq.com/ldls-laser-driven-light-source>

Parameter	EQ-99XFC	EQ-77
Spectral range	170–2100 nm	200–1100 nm
Radiance	> 1 mW/nm @ 400 nm	> 0.5 mW/nm
Stability	< 0.2% RMS	< 0.5% RMS
Housing	Fused silica (SiO ₂)	Fused silica
Discharge type	Laser-sustained Xe plasma	Laser-sustained
Arc spot size	< 100 μm	< 100 μm
Output	SMA-905 fiber	SMA-905 fiber

★★★★★ LITERAL SPEC MATCH — Excelitas Technologies — CERMAX

Product: PE300BF / PE150BF / PE75BF

CERMAX = Ceramic Maximum. The Al₂O₃ ceramic reflector IS the dielectric housing — Xe arc burns inside a parabolic ceramic cavity producing collimated UV-Vis-NIR output. Long life (1000+ hrs), cold-restrike capable. Available 75 W–1600 W.

Website: <https://www.excelitas.com/product-category/cermax-xenon-arc-lamps>

Model	Power	Life	Range	Color temp
PE75BF	75 W	500 hr	240–2500 nm	5500 K

PE150BF	150 W	1000 hr	240–2500 nm	5800 K
PE300BF	300 W	1000 hr	240–2500 nm	6000 K
PE1000	1000 W	500 hr	230–2500 nm	6200 K

★★★★ CLOSEST 528 nm NATURAL PEAK — Topanga Technologies / Luxim — Sulfur Plasma

Product: LiFi Sulfur Plasma Lamp

Sulfur vapor microwave plasma in quartz dielectric bulb. Natural emission peak ~520 nm — closest of any discharge lamp to 528 nm SPR target without bandpass filtering. 430–750 nm visible band, electrodeless, 50,000+ hr life.

Website: <https://www.topangatech.com>

Parameter	Value
Peak emission	~520 nm (natural)
Spectral range	430–750 nm
Drive	2.45 GHz microwave
Housing	Fused quartz dielectric bulb
Lifetime	50,000+ hours
Input power	1000–1400 W

★★★★ — Newport / MKS Instruments — Oriel Series

Product: Model 66902 / 69911 Xe Arc

Industry-standard Xe short arc systems, quartz envelope, 150 W–1000 W, 200–2500 nm continuum. The 66902 (150 W) is a workhorse for photocatalysis research.

Website: <https://www.newport.com/c/arc-lamp-sources>

Model	Power	Range	Notes
66902	150 W	200–2500 nm	Photocatalysis / SPR standard
69911	300 W	200–2500 nm	High irradiance
Hg-Xe 66142	200 W	185–2500 nm	UV-enhanced + 546 nm line

★★★★ — Hamamatsu Photonics

Product: L10671 / L9455 Flash Xe

Xe short arc lamps (CW and flash) plus fiber-coupled output. L9455 flash series provides pulsed μ s pulses for time-resolved PGM/REE luminescence.

Website: <https://www.hamamatsu.com/us/en/product/light-sources/xenon-flash-lamps.html>

Model	Type	Range	Power
L10671	CW Xe	200–2000 nm	150 W
L11035	CW Xe	200–2000 nm	75 W compact
L9455	Flash Xe	200–1000 nm	pulsed < 10 ms

★★★ — ams OSRAM — XBO Series

Product: XBO 75W / 150W / 300W

Ultra-high pressure Xe short arc, sub-mm arc gap, extremely high luminance. UV-transmitting quartz, output to 230 nm — covers Pt (265 nm), Os, Ir primary lines.

Website: <https://www.ams-osram.com/products/lamps/xenon-short-arc-lamps>

Model	Power	Arc gap	Color temp
XBO 75W/2	75 W	0.8 mm	6200 K
XBO 150W/4	150 W	1.2 mm	6200 K
XBO 300W/4	300 W	2.0 mm	6000 K
XBO 1000W/HS	1000 W	5.5 mm	6100 K

★★★ — Ushio America**Product: UXL / SX Series**

Xe short arc and Hg-Xe, 230–2500 nm, ceramic holders for harsh environments.

Website: <https://www.ushio.com/products/arc-lamps/>

Model	Power	Range
UXL-75XE	75 W	230–2500 nm
UXL-300D	300 W	230–2500 nm
UXL-500D	500 W	230–2500 nm

★★★ — Resonance Ltd — Canada**Product: Microwave Electrodeless Lamps**

Electrodeless microwave-excited discharge lamps, gas-fill selectable (Xe, Kr, Hg, Ar). Quartz dielectric bulb, 2.45 GHz driver, pulsed or CW. Widely used in analytical spectroscopy.

Website: <https://www.resonance.on.ca>

Fill gas	Key emission lines
Xenon	823, 828, 881 nm + continuum
Mercury	185, 254, 365, 405, 436, 546, 578 nm
Krypton	758, 760, 769, 811 nm

★★★ — Edinburgh Instruments**Product: Xe900 / μ F920H Xe Flashlamp**

Pulsed Xe flashlamps (< 1 μ s pulses) for time-resolved luminescence of Eu³⁺■, Tb³⁺■, Nd³⁺■ during REE separation monitoring. 200–900 nm, quartz housing.

Website: <https://www.edinst.com/products/light-sources/>

Model	Type	Pulse	Range
Xe900	CW Xe	CW	200–900 nm
μ F920H	Flash Xe	< 1 μ s	200–900 nm

★★ — Heraeus Noblelight**Product: DBD Excimer / Xe Flash Lamps**

True dielectric barrier discharge (DBD) excimer lamps: 172 nm Xe⁺, 222 nm KrCl^{*}, 308 nm XeCl^{*} — narrow UV bands for Pt/Os/Ir primary lines. Also broad-spectrum Xe flash.

Website: https://www.heraeus.com/en/hng/products_and_solutions/uv_curing/xenon_flash_lamps.html

Type	Wavelength	PGM application
Xe ⁺ excimer	172 nm	Os/Ir VUV photolysis
KrCl [*] excimer	222 nm	Ir primary line region
XeCl [*] excimer	308 nm	Rh photocatalysis
Xe flash	200–2500 nm	Broad-spectrum photocatalysis

Spectral Coverage Matrix vs PGM/REE Targets

Target	λ (nm)	Energetiq LDLS	CERMAX Xe	Sulfur Plasma	Hg-Xe	DBD
Au SPR	520–528	✓ continuum	✓	★ peak ~520	✓	—
Ag SPR	400–420	✓	✓	—	✓ 405nm	—
Pd	340 nm	✓	✓	—	✓	XeCl~308
Rh	343 nm	✓	✓	—	✓	XeCl~308
Pt	265 nm	✓	✓	—	✓	KrCl~222
Os/Ir	224–226	✓	✓	—	✓	Xe ⁺ 172
Eu ³⁺	394 nm	✓	✓	—	✓ 405nm	—
Tb ³⁺	365 nm	✓	✓	—	✓ 365nm	—
Er ³⁺	660 nm	✓	✓	✓	✓	—
Nd ³⁺	808 nm	✓	✓	—	✓	—
Yb ³⁺	980 nm	✓	✓	—	✓	—

Priority Purchase Path

Goal	Product	Manufacturer	URL	Est. Cost
Best UV-Vis-NIR coverage	EQ-99XFC LDLS	Energetiq	<link href="https://www.energetiq.com" color="#0D47A1"><u>energetiq.com</u>	\$15,000
Closest 528 nm natural	Sulfur lamp	Topanga	<link href="https://www.topangatech.com" color="#0D47A1"><u>topangatech.com</u>	\$400
Budget Xe + filter	PE150BF CERMAX	Excelitas	<link href="https://www.excelitas.com/product-category/cermax-xenon-arc-lamps/" color="#0D47A1"><u>excelitas.com</u>	\$500
Pulsed REE time-resolve	µF920H Flash	Edinburgh	<link href="https://www.edinst.com/products/light-sources/" color="#0D47A1"><u>edinst.com</u>	\$2,000
Deep UV Pt/Os/Ir	Xe ⁺ 172nm DBD	Heraeus	<link href="https://www.heraeus.com" color="#0D47A1"><u>heraeus.com</u>	\$1,000

All hyperlinks active — tap/click to open. Cross-referenced: STARE Research Division PDF "High-Capacity LED Systems for Metallurgical Photonic Processing" June 25, 2026.